

CS 4134 – Quantum Computation and Information Processing

Exam 3

Virginia Tech Department of Computer Science, Spring 2026
Instructor: Sumeet Khatri

Name:

Date: Thursday, March 26, 2026

Number of problems: 6

Time allowed: 75 minutes

Instructions: Please show all of your work. Your grade will be based not only on writing down the correct answer, but having a complete solution.

Grading: You will start from a grade of 100%. For every missed step, unclear or unexplained step, and mistake, you will lose two points, with a maximum of 10 points of deduction per problem. Unattempted problems result in a 10-point deduction.

Potentially helpful definitions

- Pauli matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

- Pauli- X basis vectors: $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$.

- Bell state vectors:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0, 0\rangle + |1, 1\rangle), \quad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|0, 0\rangle - |1, 1\rangle),$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|0, 1\rangle + |1, 0\rangle), \quad |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|0, 1\rangle - |1, 0\rangle).$$